Reproducing Evolutionary Suicide in the Life Engine

Oct 16, 2026 · @Ashley

Summary

The unmodified Life Engine produced clear evolutionary suicide; our modified savanna made suicide possible but harder, and it produced failure to adapt and predator boom-bust instead, including in 18 follow-up runs of up to 200,000 ticks with less rain, more leopards or stronger fire.

- **Base engine, 18 runs:** in all 3 seeds of the plant-kill control, zebras seeded without a killer cell evolved one. The trait spread, the population boomed to 307 to 560, the killers destroyed every grass and bush plant, and the zebras went extinct between ticks 5,900 and 14,900. That meets all four of our evidence criteria.
- **Modified engine, 79 runs** (5 main runs of 100k ticks, 30 ablation runs, 36 sweep runs and 8 mechanism runs): leopards boomed and then died out in every main seed (peak 26 to 35, extinct by tick 3,600 to 12,800). Rhinos collapsed in every seed. Zebras survived when they evolved a streamlined grazing body (a second mouth, and no eyes, kick or stripes they no longer needed) and faded out by failure to adapt when they did not.
- **Environment mattered as much as traits:** removing the river or the soil-water system killed every zebra lineage by tick 40k, compared with 1 in 5 in the full design. Removing stripes made zebras die out fastest. Removing the cache saved every zebra lineage, because leopards died sooner.

Question and evidence rule

We asked whether a savanna of zebras, rhinos, leopards, grass and bushes can drive itself extinct by mis-adaptation, and how that differs from failure to adapt.

- **Evolutionary suicide:** a trait spreads because it helps individuals, and the same trait removes something the population depends on, so the population collapses.
- **Failure to adapt:** the species lacks the traits it needs and declines without first succeeding.

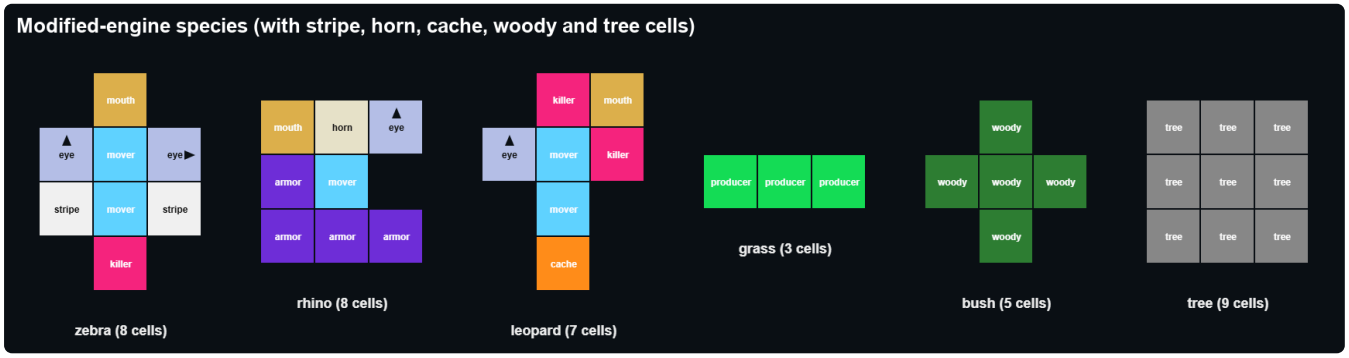
As we committed to in Assignment 3, a decline alone does not count as suicide. The analysis script (`analyze.py`) labels a collapse as evolutionary suicide only when all four of these hold:

1. The population grew to at least 1.5 times its starting size.
2. A trait spread before the decline: mouths per animal rose by 0.4 or more, or the share with a killer cell rose by 0.3 or more.
3. The food plants (grass plus bush organisms) fell by at least 25% between the peak and extinction.
4. The decline was self-caused (starvation, old age without replacement, or killing by the lineage's own evolved killers) rather than caused by another species.

Three of four with growth and trait spread is reported as a suicide candidate. Collapses without growth are failure to adapt. Collapses where other species did most of the killing are collapse under predation. For leopards we separate boom-bust from over-predation (the prey also crashed) and boom-bust from a cohort ageing out.

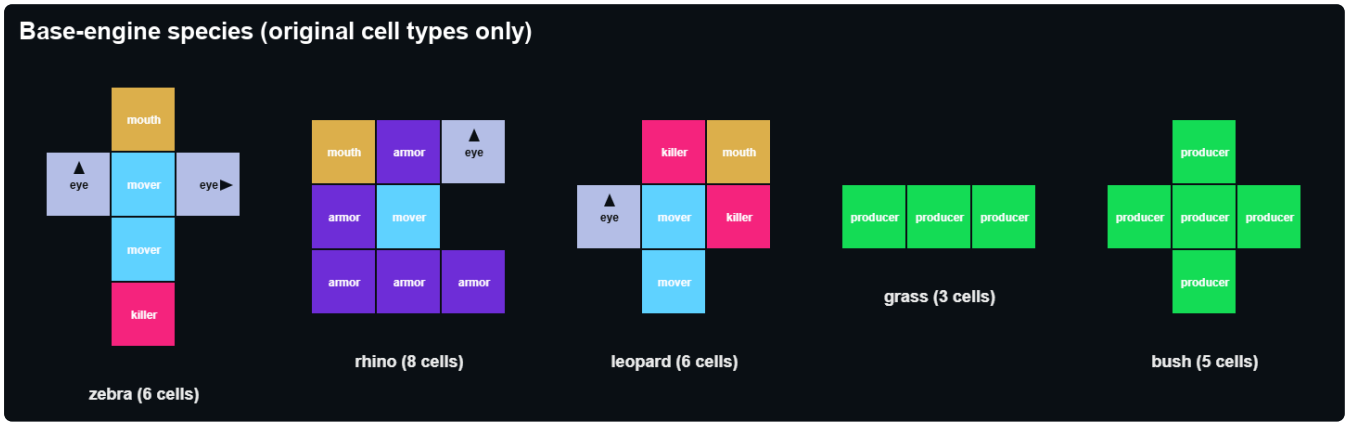
Species

Each species copies a real animal or plant's strengths and weaknesses, so that every modification acts on a trade-off the species already has. The base runs use the same body plans with the new cells removed or swapped for original ones (the horn becomes armour, and stripes and the cache are dropped).



Species	Cells (modified engine)	Real-world basis	Strengths	Weaknesses
Zebra	2 movers, 1 mouth, forward eye, side eye, rear kick (killer), 2 stripes (8)	Fast, vigilant herd grazer that kicks predators	Sprints at 2 steps per tick when chasing or fleeing; herd camouflage; kick wounds leopards	One mouth; poor on bush food (0.35 value); no armour

Species	Cells (modified engine)	Real-world basis	Strengths	Weaknesses
Rhino	1 mover, 1 mouth, horn, forward eye, 4 armour (8)	Huge, armoured, slow-breeding grazer	Horn is immune to killers and gores attackers; armour adds 3 health per cell (20 health in total)	Slow; armour costs 2 extra food per calf (16 food per calf); poor on bush food
Leopard	2 movers, front killer and side killer beside the mouth (jaws), side eye looking forward, cache (7)	Stealthy ambush hunter that hoists kills into trees	Sprint; bite does 4 damage; eats its kill on the spot; can bank surplus food near trees	Eats meat only; no armour; gored by horns, kicked by zebras
Grass	3 producers in a row	Fast-spreading, shallow-rooted, grazed to the ground	Reproduces fastest; drops full-value food	Uses a lot of water; mouths can bite the plant itself
Bush	5 woody cells in a plus	Woody, thorny, deep-rooted	Cannot be grazed; uses half as much water	Its browse feeds grazers poorly
Tree (modified only)	3x3 tree cells	Shade and caching sites	Lives 6 times longer; resists fire; hides animals beside it	Very slow; thirsty



Starting numbers in every run: 140 grass, 45 bush, 14 trees (modified only), 36 zebras in 6 herds, 12 rhinos and 16 leopards, on a 160 x 100 grid. Zebras start without a flight response; fleeing can evolve through brain mutation.

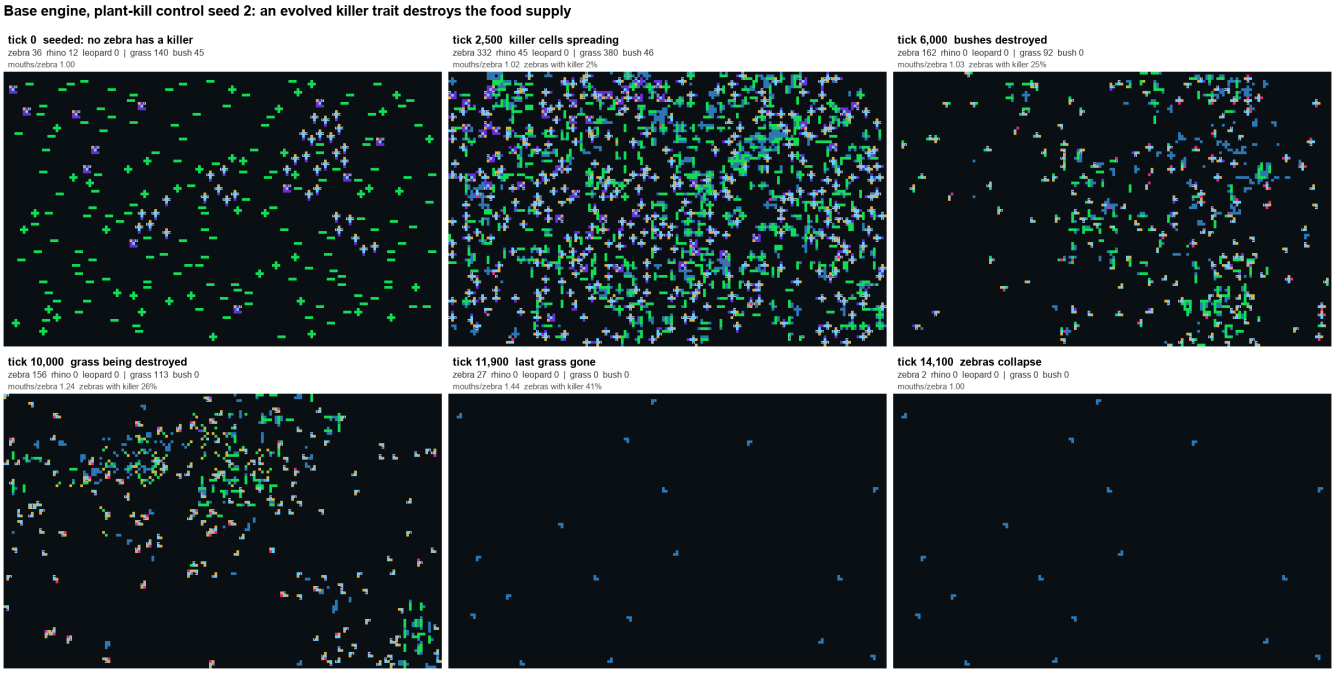
Base engine: suicide by an evolved plant-killer

In the unmodified engine, zebras that evolved killer cells destroyed their own food plants and went extinct in 3 of 3 seeds. That is the clearest evolutionary suicide in the project.

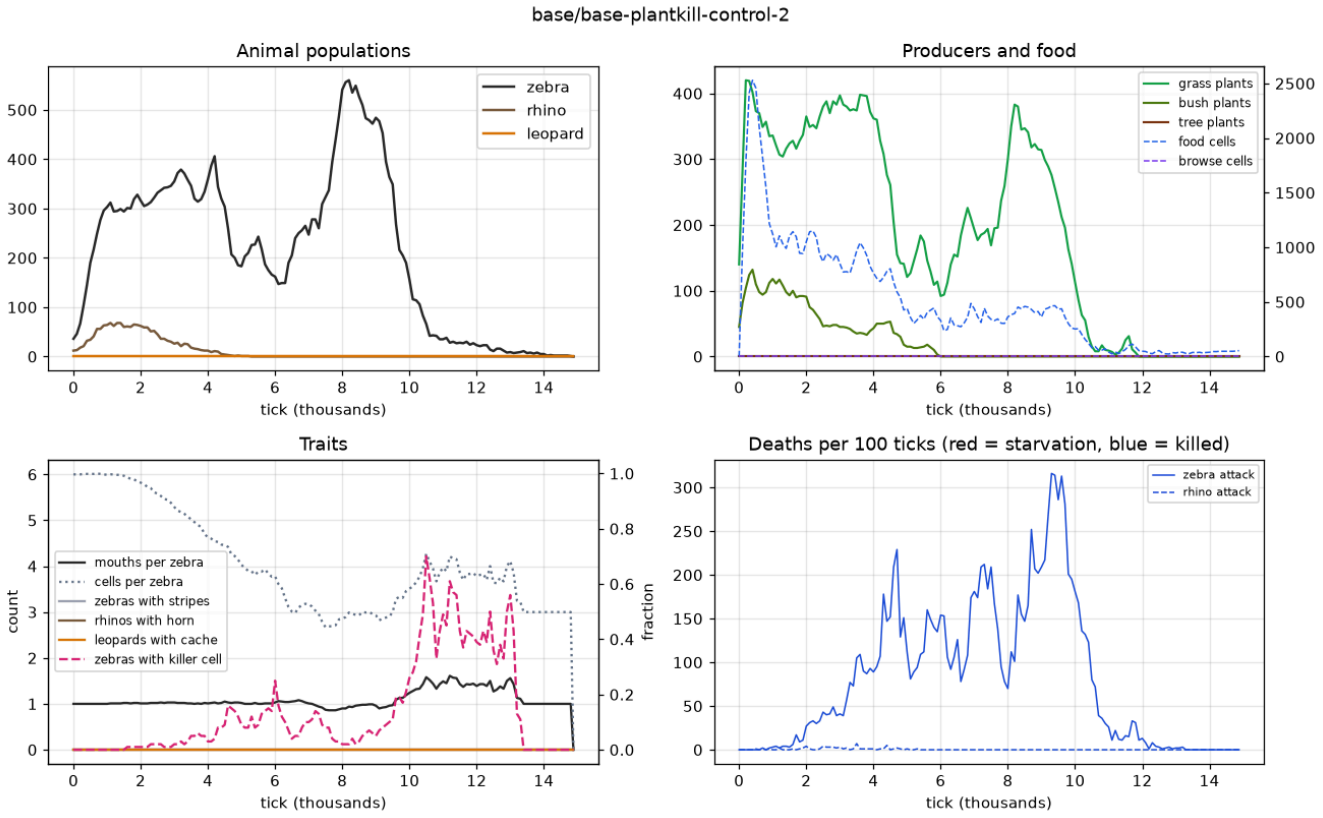
The base runs use the original rules only. Every new setting stays at a default that reproduces the original engine, and mutation cannot create the new cell types. Each condition ran for 3 seeds of 40,000 ticks.

Why the base engine allows this route

In the original engine, a killer cell damages anything it touches, plants included, and a dead plant turns into food. A zebra whose killer cell destroys a grass plant therefore gets a windfall of food and breeds faster, so the trait spreads. Every kill also removes a producer for good. Once the plants are gone, nothing makes food, and the zebras die of old age without replacing themselves.



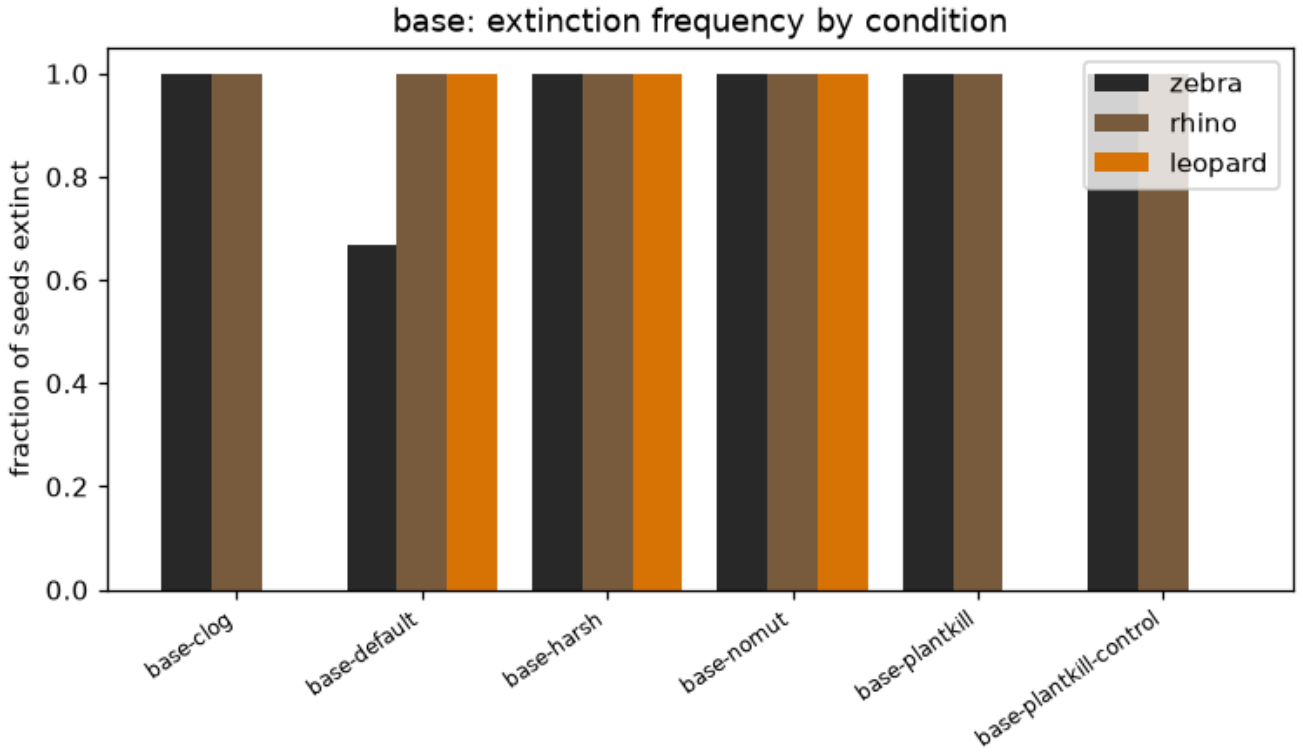
In the plant-kill control (insta-kill on, killers reach all 8 neighbours, food production 3), zebras started **without** a killer cell. In seed 2, one evolved and spread through the herd. The population climbed to 560 by tick 8,200. Bushes were gone by tick 6,000 and the last grass by tick 11,900. Zebras collapsed at tick 14,100 and were extinct by tick 14,881. Seeds 1 and 3 followed the same sequence faster (peaks of 307 and 312, extinct at ticks 6,113 and 5,940).



All base conditions

Condition	What it tests	Zebra outcome (3 seeds)	Rhino outcome	Leopard outcome
Plant-kill control	Killer trait must evolve	Evolutionary suicide x3 (all four criteria)	Extinct x3	Not seeded
Plant-kill (killer seeded)	Same route, trait given at the start	Extinct x3 by about tick 4,100; fails the trait-spread criterion because the trait was seeded, not evolved	Extinct x3	Not seeded
Default settings	Original engine as shipped	Extinct x2 (median tick 21,150), survived x1	Extinct x3	Boom then bust x3 (peak about 32, extinct by about tick 4,100)

Condition	What it tests	Zebra outcome (3 seeds)	Rhino outcome	Leopard outcome
Harsh (food production 2)	Scarcity	Extinct x3 (median tick 4,500)	Extinct x3; in seed 2 rhinos evolved killers and met all four criteria (peak 313, extinct at tick 30,452)	Boom then bust x3
No mutation, harsh	Failure to adapt baseline	Extinct x3 (median tick 6,224), with no trait change possible	Extinct x3	Boom then bust or failure x3
Clogging (food production 10, grows into 8 neighbours, few grazers)	Plants choking the world	All animals extinct by about tick 3,500	Extinct x3	Not seeded



What the base engine cannot show

- Grass and bush drop identical food, so the hypothesis that zebras switch to worse bush food cannot happen.
- Nothing starves. Extinction only happens through failure to breed before old age, so scarcity acts slowly.
- Leopards can eat plant food, and carcasses become plant food, so predators never depend on prey.
- Mouths eat only loose food and cannot hurt plants. More grazing frees space and increases plant production, so overgrazing is impossible. The only way to destroy plants is with killer cells, which is why the suicide above runs through killers rather than mouths.
- In the clogging runs, grass filled about 2,000 organisms' worth of space and births became impossible. That is an environmental artefact of food blocking reproduction, not mis-adaptation.

Modifications

Every modification opens a route to collapse and also adds an escape from it, so collapse is possible but never automatic. All of them default to off, so the base runs reproduce the original engine.

Member features

Feature	Owner	Cell or file	Opens a route to collapse	Escape that makes it hard
Horn cell	Vimbai	`HornCell.js`	Horned rhinos trample nearby plants and food (5% per tick per neighbour)	Horn is immune to killers and gores attackers (50% per tick), so rhinos survive predation
Stripe cell	Moreblessing	`StripeCell.js`, `EyeCell.js`	More surviving zebras put more pressure on grass	A leopard's eye misses a striped zebra with probability 25% per striped neighbour within 4 cells, up to 85%

Feature	Owner	Cell or file	Opens a route to collapse	Escape that makes it hard
Cache cell	Ashley	`CacheCell.js`, `Organism.js`	Banking food delays breeding	Surplus (at least half a litter's food) is banked, up to 3 units per cache cell, near a tree; it is spent automatically when hunger would kill
Soil water and fertility	Fourth member	`ProducerCell.js`, `WoodyCell.js`, `WorldEnvironment.js`	Grass needs water to produce (0.5 per event), so overgrazed, dry soil recovers slowly	Rain refills soil (0.2% per tick towards a Perlin-noise capacity); deep-rooted bushes use half as much water and expand when grass declines

Shared ecology needed to test the hypothesis

Change	Why it was needed
Bush food (browse) worth 0.35 of grass food	Gives zebras a worse fallback food, the core of our hypothesis
Grazing: a mouth next to a grass plant damages it (2% per tick)	Lets mouths deplete plants, which the base engine cannot do
Hunger: pay 1 food every 80 ticks; die after 10 missed payments in a row	Makes scarcity lethal
Carcasses become meat; jaws (a killer beside a mouth) digest meat only; a leopard eats its kill on the spot	Makes leopards depend on prey
Sprint: extra mover cells add a step per tick, but only when chasing or fleeing	Lets predators catch prey without fast animals trampling their own food
Armour costs 2 food to grow and adds 3 health	Makes rhinos tough but slow to breed, as real rhinos are

Change	Why it was needed
Weapons (a killer away from the mouth, like the zebra kick) do 2 damage and only strike predators	Stops zebra kicks from killing rhinos (found and fixed during testing)
River: impassable 3-cell channel; only animals of 9 or more cells can wade; riverbank soil refills 3 times faster	Permanent green refuge; splits the map
Trees: long-lived 3x3 plants that hide nearby animals from the other diet (60%)	Cover for prey and ambush for leopards; the only place caches fill
Fire: random fires (more likely in the dry season) spread through dry vegetation; wet ground and gaps stop them	Grazed land burns less, so diversity changes the environment
Eco logger (`EcoStats.js`) and harness	Records population, traits, food, water and deaths by cause every 100 ticks, and saves screenshots at fixed ticks and at every collapse

How the features connect

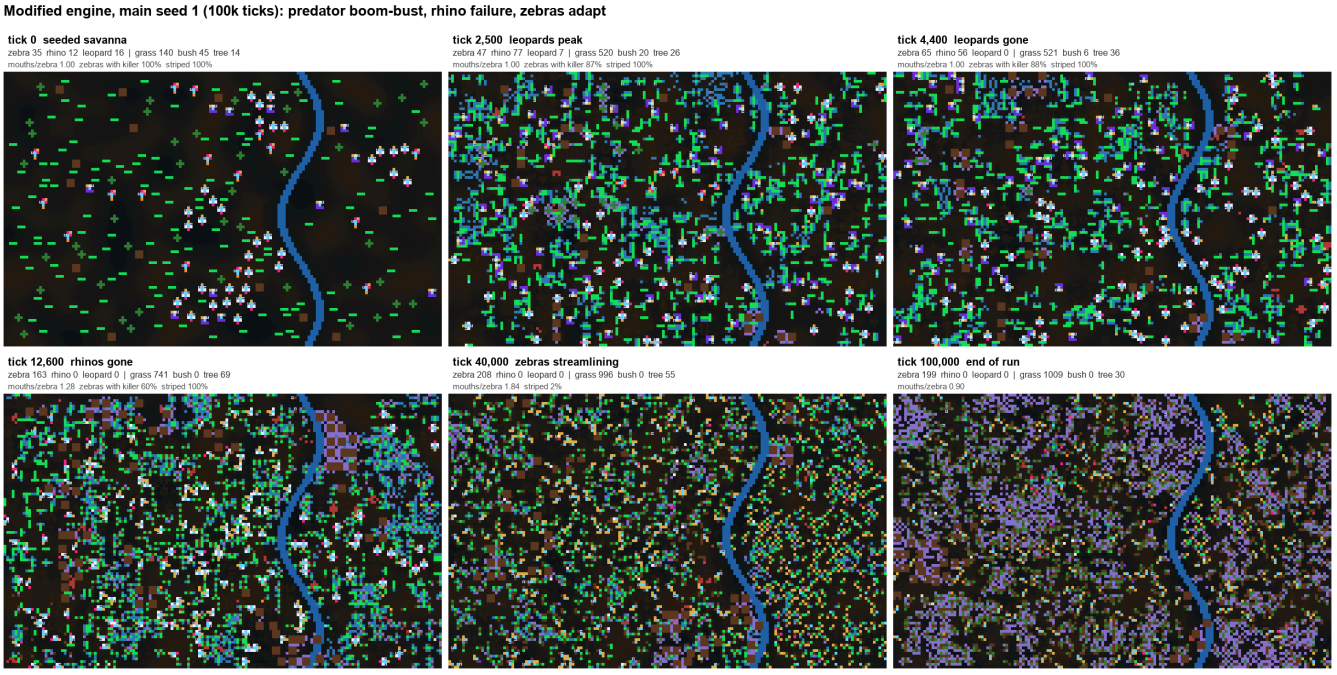
1. Rain and soil water set how fast grass regrows. Zebras and rhinos graze it, and trampling by horned rhinos adds to the damage.
2. When grass falls, water is freed for bushes, which expand and offer worse food.
3. Leopards hunt zebras and rhinos. Stripes, tree cover, the kick and the horn reduce kills, and the cache lets leopards wait out lean periods.
4. The river keeps a strip of wet, fertile ground that only rhinos can reach across. Fires remove dry vegetation, and grazed ground acts as a firebreak.

No hard-coded outcomes

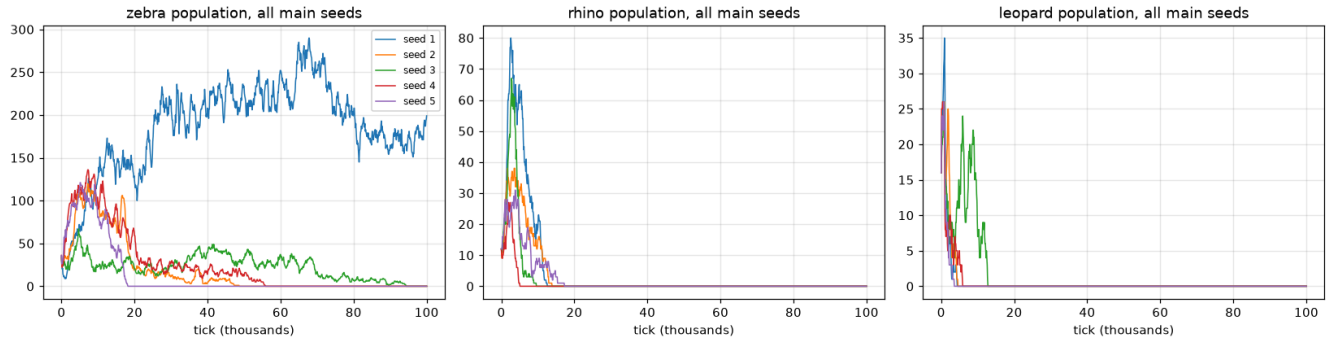
No line of code sets a population, a death or a survival. Every death comes from the same per-organism rules: old age, damage from cells, or hunger. The variant worlds only change starting organisms and settings (`savanna-experiments/make-world.js`). Outcomes differ between seeds of the same condition, which a scripted result could not do.

Modified engine: five 100,000-tick runs

In all five seeds, leopards boomed and died out within 13,000 ticks and rhinos faded out within 18,000. Zebras survived only where they evolved a second mouth.

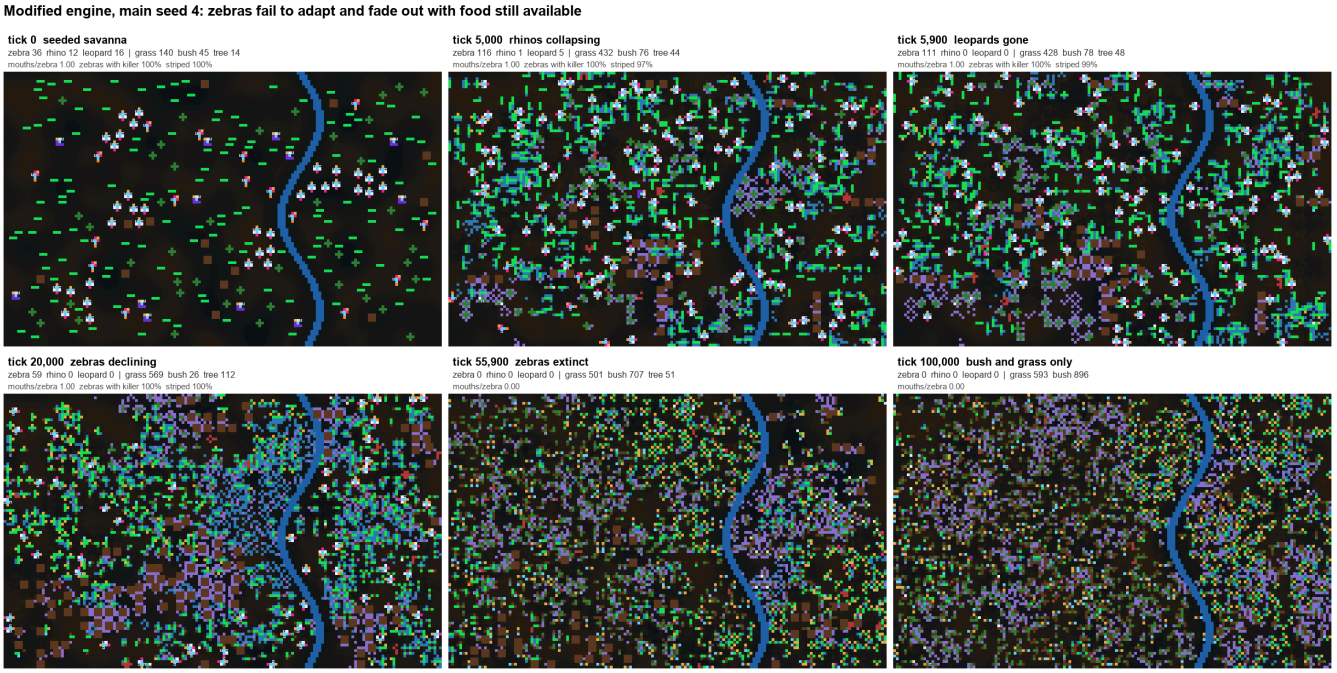


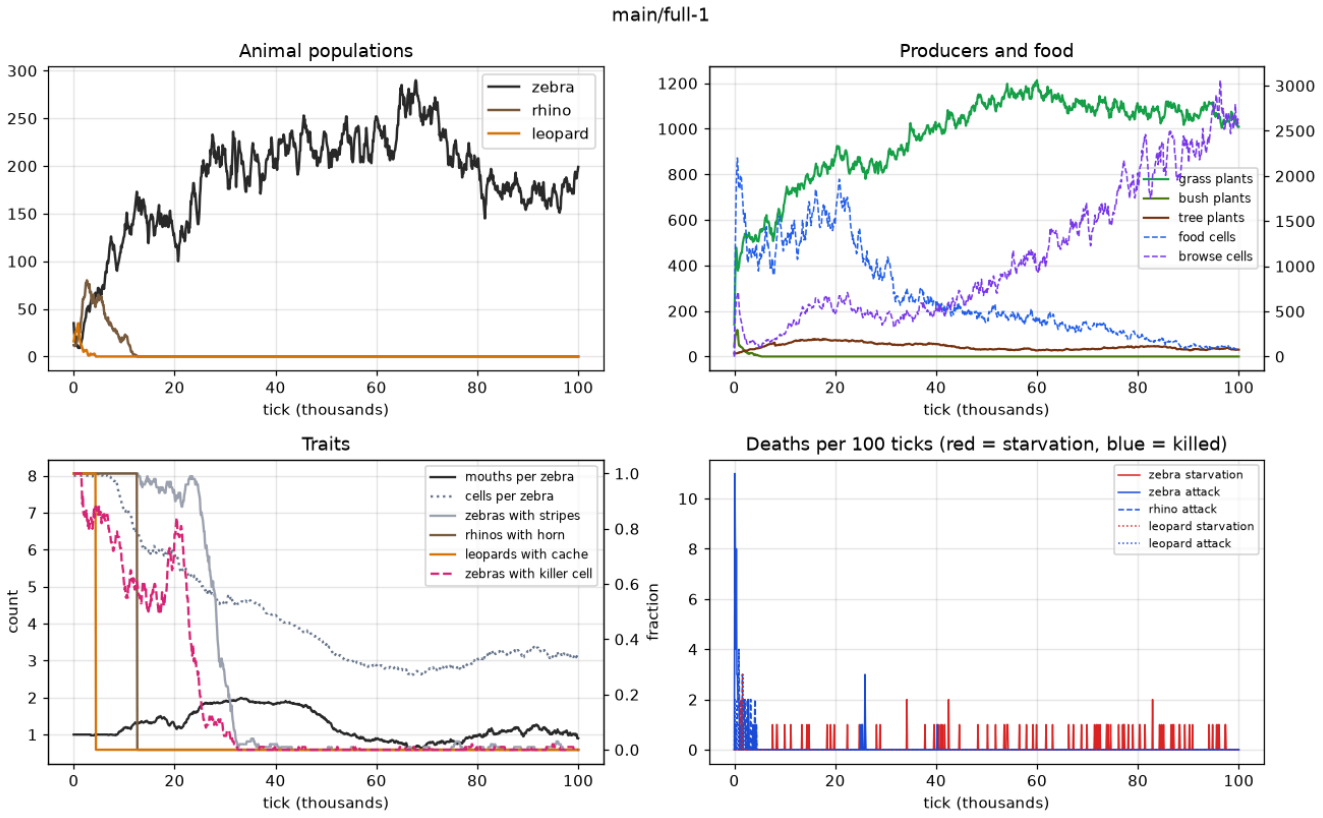
Seed	Leopards: peak, extinct at tick	Rhinos: peak, extinct at tick	Zebras: peak, outcome	Max mouths per zebra
1	35, 4,400	80, 12,600	290, survived (199 at tick 100k)	1.99
2	26, 4,600	38, 13,900	120, extinct at 48,700	1.10
3	26, 12,800	67, 9,600	67, extinct at 94,400	1.20
4	26, 5,900	27, 5,100	136, extinct at 55,900	1.00
5	26, 3,600	31, 17,300	121, extinct at 18,200	1.03



What happened to each species

- **Leopards: boom then bust.** Every seed began with 16 leopards and they rose to 26 to 35 within about 1,000 ticks by eating the herd (57 to 144 zebras and 17 to 66 rhinos killed per run). They then died out. Old age was the main cause in seeds 1 and 3, as the first boom generation aged out without enough kills to replace itself. Zebra kicks were the main cause in seeds 2 and 4 (36 and 48 leopards killed). Caches were used 76 to 870 times per run, which delayed but never prevented extinction. Zebras did not crash before the leopards died, so this is a predator that failed to persist rather than over-predation.
- **Rhinos: failure to adapt.** Rhinos rose from 12 to 27 to 80, then died out. Nearly all deaths were old age (54 to 358 per run) with only 17 to 66 killed by leopards. At 16 food per calf they could not replace themselves once zebras took most of the grass. No horn-trampling collapse appeared.
- **Zebras: survival depended on adapting.** In seed 1, mouths per zebra rose from 1.0 to 1.99 and the herd stayed large. In seeds 2 to 5 the mouth trait did not spread. Zebras then declined slowly over tens of thousands of ticks, dying of old age (1,081 to 2,029 deaths) while food was still on the map. Starvation killed only 4 to 81 per run.
- **Plants and fire:** grass grew to between roughly 500 and 1,200 plants, and bushes expanded or vanished depending on how much grass took their water. Fires were frequent (29 to 41 per run) but small (154 to 381 cells burnt in total), so fire was a minor force.

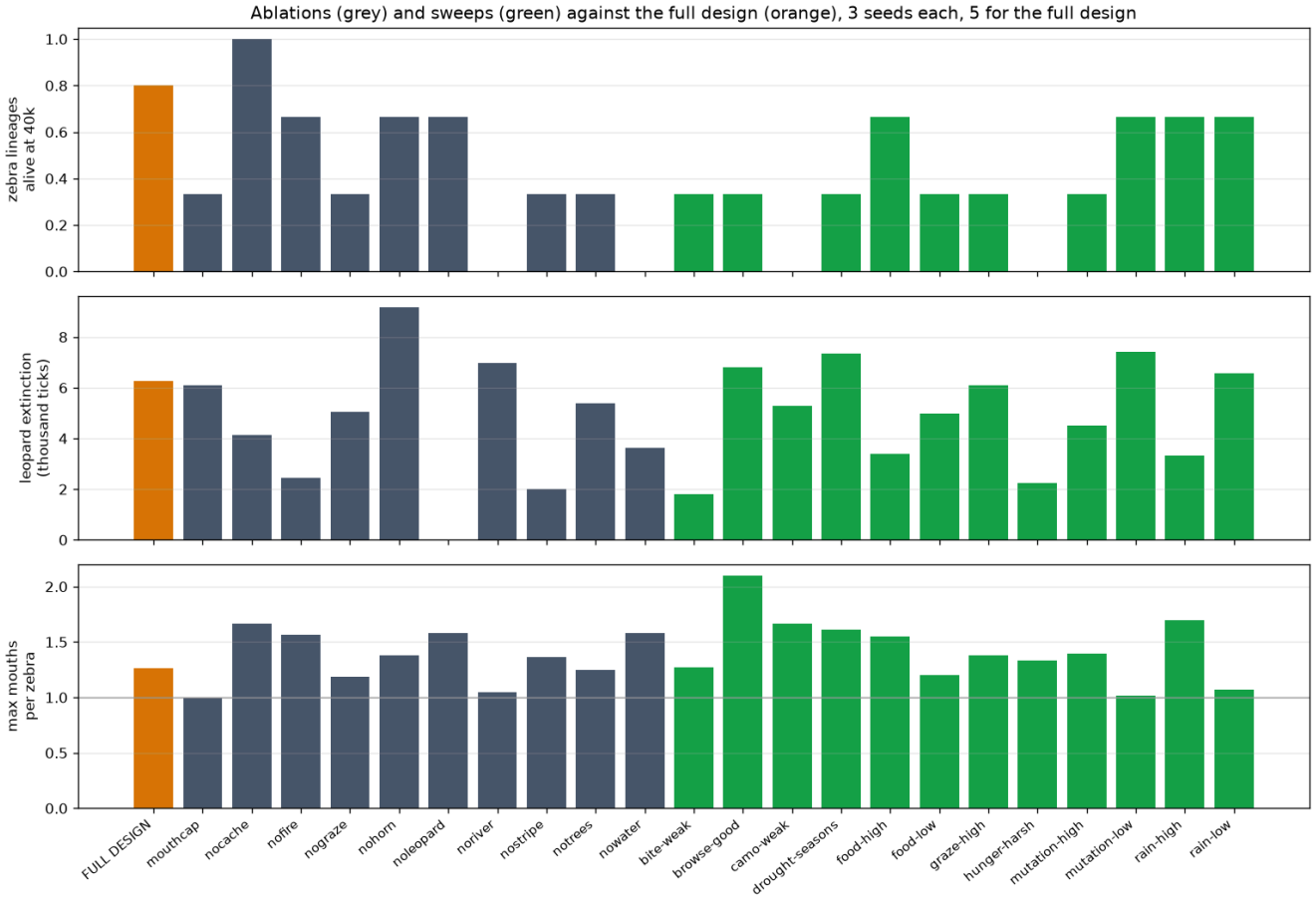




Ablations and sweeps

The river, soil water and stripe camouflage were what kept zebras alive. Without the cache, leopards died sooner and every zebra lineage survived.

Each ablation switches off one feature, and each sweep changes one setting of the full design. Each ran for 3 seeds of 40,000 ticks. The reference is the five main runs read at tick 40,000, when 4 of 5 zebra lineages were alive. With 3 seeds, a one-seed difference is within noise. We treat only 0 of 3 or 3 of 3 as a clear signal.



Condition	Zebra lineages alive at 40k	Other notable result	Reading
Full design (reference)	4 of 5	Leopards extinct by about tick 6,300 on average	Baseline
No river	0 of 3 (median extinction tick 18,300)	Lowest zebra peak (84)	The riverbank refuge kept zebras going
No soil water	0 of 3 (median tick 21,100)	Grass spreads unchecked	Water limits matter to zebras; the exact cause of their decline here was not isolated
No stripes	1 of 3 (median tick 11,300, the earliest)	Zebra peak 78	Herd camouflage protects zebras
Weak camouflage (maximum 30%)	0 of 3	Confirms the stripe result	Herd camouflage protects zebras

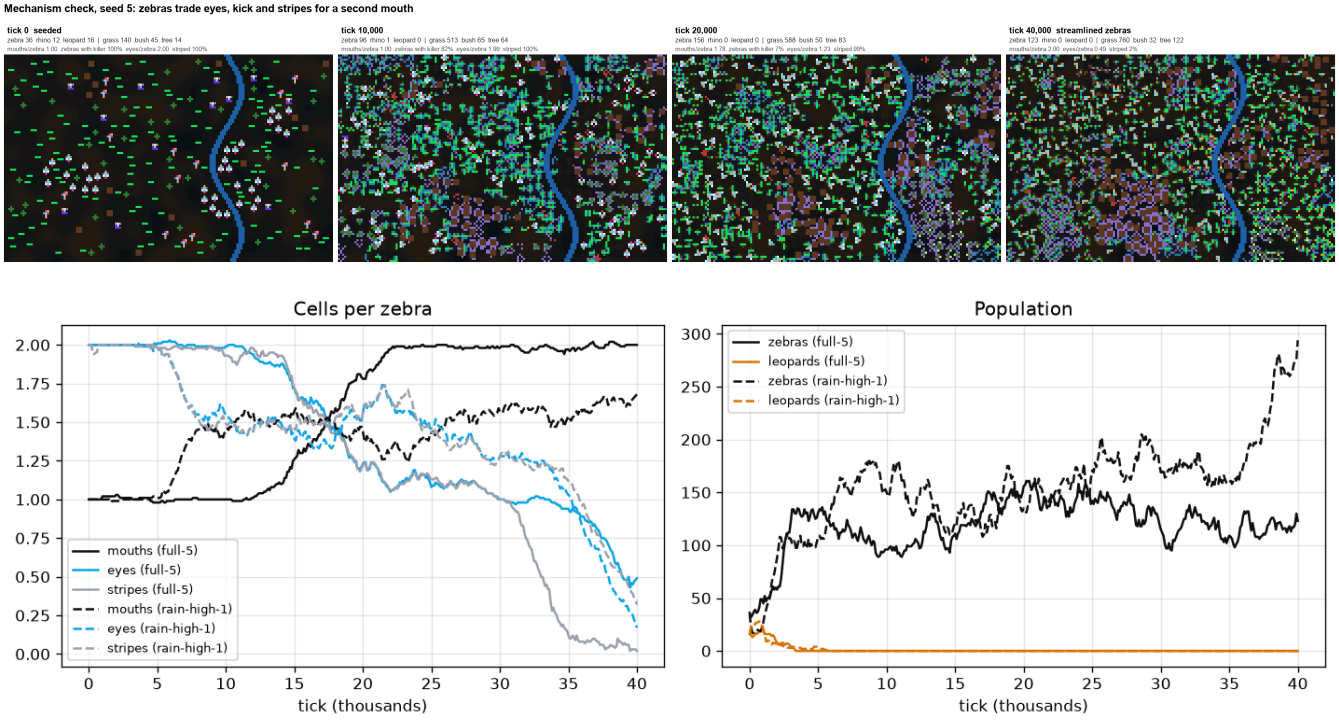
Condition	Zebra lineages alive at 40k	Other notable result	Reading
No cache	3 of 3	Leopards extinct sooner (about tick 4,100)	The cache prolongs predation, which costs zebras
No trees	1 of 3	Leopards (who could then cache anywhere) peaked at 110, and persisted in one seed	Tying the cache to trees limits leopards
No horn	2 of 3	Leopards lasted longest (about tick 9,200)	Horns shorten predator survival
Mouth cap (zebras limited to 1 mouth)	1 of 3	Mouths stayed at 1.00	Blocking the extra mouth harmed zebras: the trait is adaptive, not suicidal
No grazing	1 of 3	-	Grazing damage was not what killed zebras
No leopards	2 of 3	Zebra peak 201	Predators are not needed for zebras to fade
No fire	2 of 3	-	Fire is a minor force
High food production (8)	2 of 3	Highest zebra peak (225)	More food helps but does not guarantee survival
Low food production (3)	1 of 3	Rhinos survived in one seed	Scarcity favours slow, armoured rhinos
High or low rain	2 of 3 each	High rain gave mouths 1.70	Rain alone did not decide the outcome
Seasonal drought	1 of 3	-	Within noise
High grazing damage	1 of 3	-	Within noise
Good bush food (0.7)	1 of 3	Highest mouths (2.10)	A better fallback did not rescue zebras

Condition	Zebra lineages alive at 40k	Other notable result	Reading
Harsh hunger (6 missed payments)	0 of 3	Leopards extinct by about tick 2,200	Energy stress kills grazers and predators alike
Weak bite (damage 2)	1 of 3	Leopards extinct by about tick 1,800	Leopards need a strong bite to survive at all
High or low mutation	1 of 3 and 2 of 3	Low mutation kept mouths at 1.01	Faster change did not help zebras

Mechanism check: zebras streamline

When leopards disappear, zebras that survive swap costly defensive cells for a second mouth. This is an adaptation, not a suicide.

The main logs recorded mouths but not other cell types, so we added per-cell counts and reran 8 runs (5 seeds with high rain, 3 with default settings) with screenshots every 5,000 ticks.

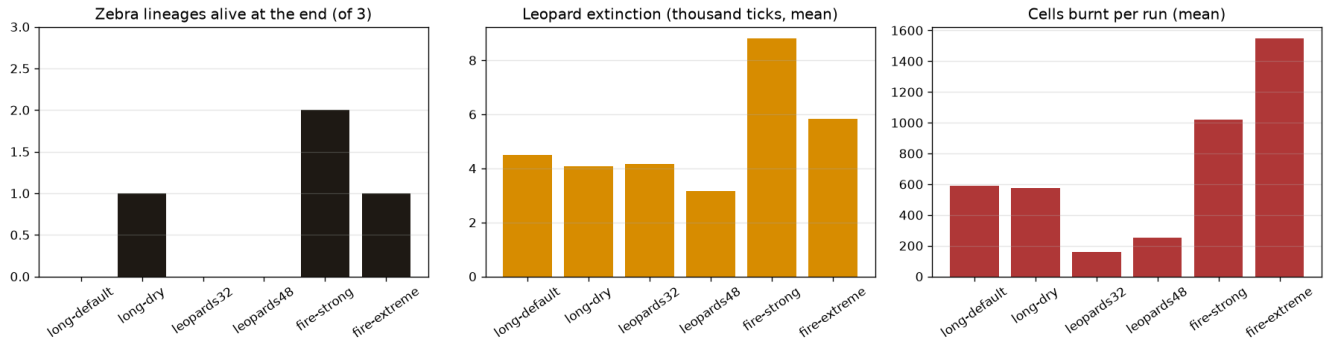


- In default seed 5, eyes per zebra fell from 2.0 to 0.49, stripes from 2.0 to 0.02, and the kick from 1.0 to 0 by tick 25,000. Mouths rose to 2.0, and the herd held at about 110 to 150.

- In high-rain seed 1, the same shift took the body from 8 to 4.3 cells, and the herd grew to 294 by tick 40,000.
- Once leopards are gone, eyes, stripes and the kick are dead weight. A smaller body needs less food per calf, and a second mouth gathers food faster, so the streamlined lineage wins.
- In the four high-rain seeds where zebras died, the shift never started. Mouths stayed at 1.0 and zebras faded out by tick 10,000 to 30,000 with food still present. That is failure to adapt.
- Grass food did not crash under two-mouthed zebras: it stayed between roughly 500 and 2,000 cells. Water-limited regrowth plus the bush fallback kept the extra-mouth trait from removing its own food, which is the third criterion our hypothesis needed.

Follow-up runs: longer, drier, more predators, stronger fire

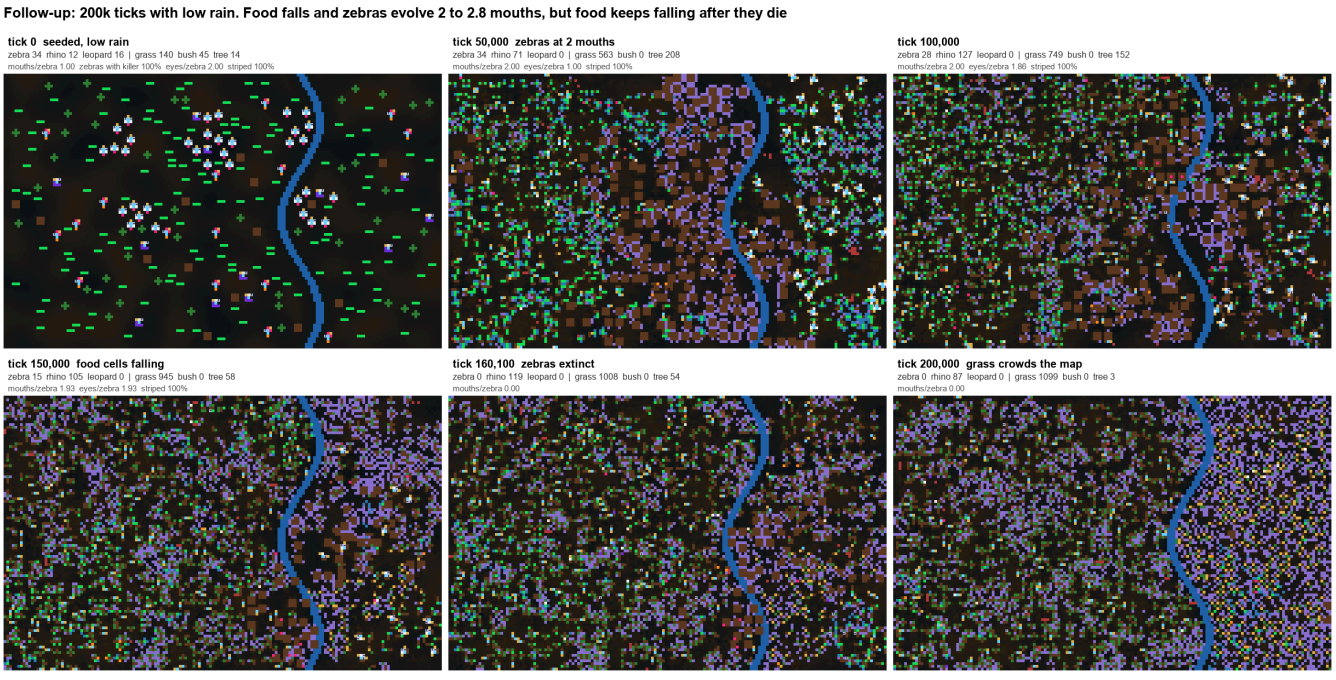
Eighteen more runs (3 seeds each) tested the three open questions. None produced zebra overgrazing suicide or a persistent predator. Stronger fire burnt more but stayed patchy.



Condition	Settings	Zebras	Leopards	Fire
Long default	200k ticks	0 of 3 survived; mouths rose to 2.0 in 2 seeds	Extinct by tick 2,400 to 7,400	About 590 cells burnt per run
Long dry	200k ticks, rain 0.0008 (40% of normal)	1 of 3 survived; mouths up to 2.8	Extinct by tick 2,000 to 7,100; in one seed the zebras also crashed (over-predation)	About 570 cells per run

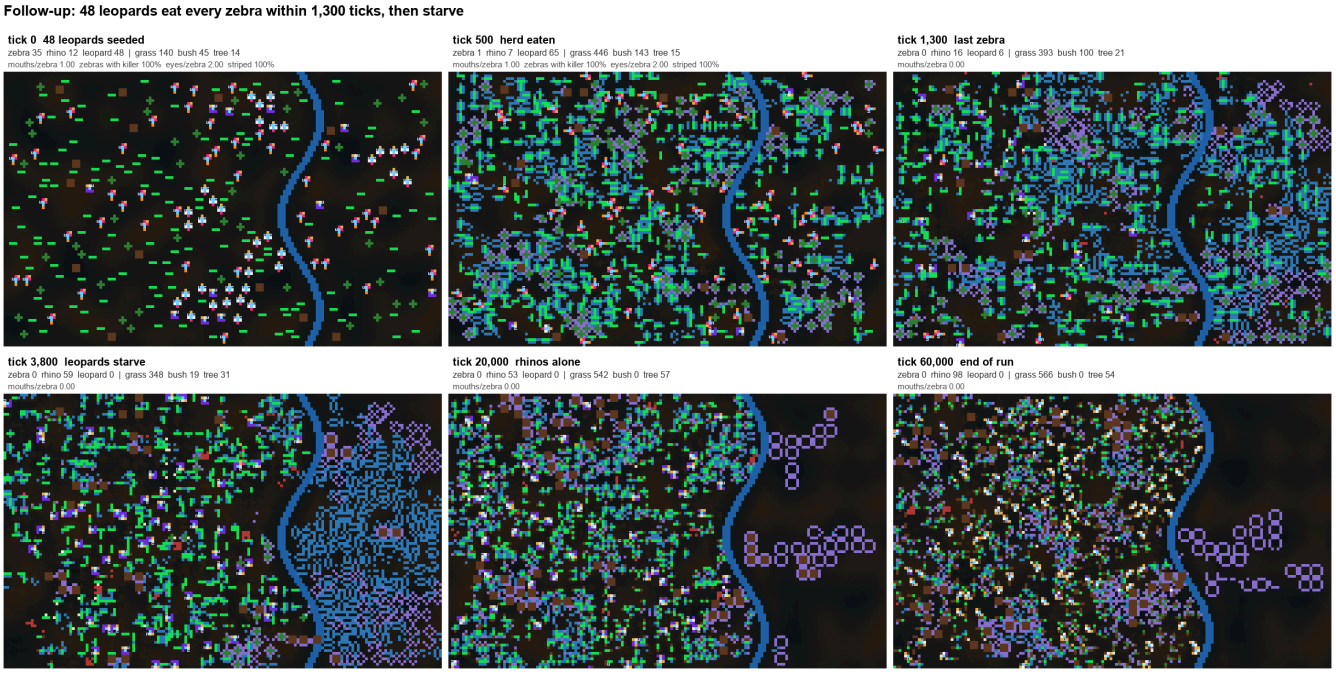
Condition	Settings	Zebras	Leopards	Fire
32 leopards	Twice the starting predators, 60k ticks	0 of 3 survived (extinct by tick 17,700 to 47,000)	Peaked at 44 to 48, extinct by tick 3,200 to 5,800	Small
48 leopards	Three times the starting predators, 60k ticks	Herd eaten within about 1,300 ticks in 2 seeds	Peaked at 66 to 70, starved by tick 2,000 to 3,800	Small
Strong fire	Fire chance 5x, spread 1.0, up to 4,000 cells	2 of 3 survived at 40k	Extinct by tick 3,500 to 15,600	About 1,020 cells per run, largest single fire 160, 270 to 340 plants killed
Extreme fire	Fire chance 10x, up to 8,000 cells	1 of 3 survived at 40k; mouths 2.0 in all seeds	Extinct by tick 3,300 to 10,900	About 1,550 cells per run, largest single fire 107, 370 to 520 plants killed

Longer and drier runs: zebras adapt to scarcity they did not cause



In long dry seed 3, grass food fell from about 1,550 cells at tick 20,000 to 166 at tick 160,000, and zebras answered by evolving 2 to 2.8 mouths. They still died out at tick 160,100. Food kept falling after the last zebra died (to 36 cells at tick 190,000), and grass plants doubled over the same period. So the food shortage came from grass crowding the map, not from zebra grazing. The extra mouth was a response to scarcity, not its cause, so this does not meet criterion 3. Doubling the run length did not change the conclusion.

More leopards: over-exploitation, not persistence



With 48 leopards the herd of 35 was gone by tick 1,300, and the leopards starved by tick 2,000 to 3,800. This is the textbook predator over-exploitation collapse: the predators destroyed their only prey and then died. It is ecological, not evolutionary. It came from the starting density we chose, with no trait spreading first, so it is not evolutionary suicide. With 32 leopards the boom was larger (peaks of 44 to 48) but ended just as fast. More predators only shortened the system's life.

Stronger fire: still limited by patchy fuel

Raising the fire chance 10 times raised burning only about 2.6 times (from about 590 to 1,550 cells per run), and no single fire passed 160 cells. Plants in this world grow as scattered small organisms on mostly wet soil, so fires run out of connected dry fuel. Fire killed hundreds of plants but did not change who survived beyond seed-to-seed noise.

Answers to the brief

1. Did an extinction event occur? Yes, in almost every run. Every leopard and rhino population in the modified main runs died out, and zebras died out in 4 of 5 main seeds and in most base conditions.

2. Mis-adaptation or failure to adapt?

- **Mis-adaptation (evolutionary suicide):** zebras in the base-engine plant-kill control, 3 of 3 seeds, and rhinos in one harsh base seed. A killer trait evolved and spread because each plant it destroyed was a meal, and the same trait wiped out the producers the population lived on. This meets all four criteria.
- **Failure to adapt:** rhinos in every modified run (slow breeding, death by old age, no earlier trait), and zebras in modified seeds where the mouth trait never spread.
- **Predator boom-bust, not suicide:** leopards boomed on the herd and then aged out or were kicked to death before the prey crashed.
- **Our main hypothesis, zebras overgrazing with extra mouths and switching to poor bush food, was not observed.** The extra mouth did spread (to 1.99 to 2.58 mouths in several runs), but grass food never fell enough to undo it. Capping zebras at one mouth made survival worse (1 of 3 lineages alive at 40k, against 4 of 5 in the full design), which shows the trait was adaptive.

3. Environment versus seeded species. The same species gave opposite results in different environments. In the base engine (no water limits, killers hurt plants, nothing starves), zebras drove themselves extinct. In the modified savanna, the same zebras either streamlined and survived or failed slowly. Within the modified savanna, removing the river or soil water was as decisive as removing a species trait (0 of 3 zebra lineages alive at 40k for each).

4. Instrumental features and traits.

- **Toward extinction:** evolved killer cells (base suicide), slow rhino breeding (armour cost), leopard dependence on a single prey, and harsh hunger.
- **Away from extinction:** river refuge, soil-water regrowth, stripe camouflage, the zebra kick (it killed leopards), and the evolved second mouth with a streamlined body.
- **Cache:** delayed leopard collapse but did not prevent it, and its main effect was to keep predation on zebras going longer.

Limitations

The strongest result (base-engine suicide) is robust across 3 of 3 seeds. The modified-engine results rest on settings we tuned by trial runs and on 3 seeds per condition.

- **Tuning history.** Before the final runs, we changed rules after seeing results, because early designs failed for mechanical reasons:
  - Leopards were redesigned three times: the killer was on the side, the eye blocked the jaws, and they could not eat their kills.
  - Sprinting was limited to chasing and fleeing, because fast zebras trampled their own food.
  - Armour cost and health were raised.
  - Weapons were limited to striking predators, after zebra kicks were found killing rhinos.

  All of these are general rules about body parts, not species switches. They were fixed before the final matrix, and every final run used the same rules.
- **Small samples.** Ablations and sweeps have 3 seeds, so a one-seed difference is noise.
- **Different run lengths.** Main runs ran for 100k ticks, ablations and sweeps for 40k. We compare them at tick 40k.
- **Not exactly repeatable.** Seeds fix the starting world but not the engine's random numbers, so reruns give different trajectories with the same overall pattern.
- **Arbitrary thresholds.** The classifier uses 1.5 times growth, a 0.4 mouth rise, a 0.3 killer rise and a 25% plant fall. The per-run evidence is in `output/summary.json` so other thresholds can be checked.
- **Lineages are labelled by their seeded ancestor.** The "grass" lineage evolved mouths (about 1 per plant by tick 100k) and a few mobile forms, but it is still counted as grass.
- **Weak fire.** Fires burnt at most a few hundred cells per run, and even 10 times the fire chance (follow-up runs) gave fires of at most 160 cells, so fire stayed a minor force.
- **Small leopard population.** With 16 leopards on a 16,000-cell map, they failed to persist in every main seed. Doubling or tripling them (follow-up runs) made them crash faster, so long-term predator-prey cycles were never observed.
- **Base engine is not completely stock.** The base runs include our Assignment 2 change that lets plants reproduce from producing food. Without it, plants without mouths never reproduce in the stock engine.

Reproducing the results

Everything is in the repository under `savanna-experiments/`, and the engine changes are under `src/`.

Step	Command or file
Build the engine	`npx webpack --mode=development`

Step	Command or file
Generate a world	`node savanna-experiments/make-world.js <variant> <seed>` (variants: `full`, `base-*`, `no*`, `mouthcap`)
Serve the engine and save outputs	`python savanna-experiments/server.py 8138`, then open `http://localhost:8138`
Run the experiment matrix	In the browser console, load `/exp/harness.js` and `/exp/jobs.js`, then call `Savanna.queue(JOBS)`
Charts and summary table	`python savanna-experiments/analyze.py` writes `output/charts/` and `output/summary.csv`
Report figures	`python savanna-experiments/make_figures.py` writes `output/figures/`
Observation log (state at every screenshot)	`python savanna-experiments/obs_log.py` writes `output/observation-log.md`

Every run folder in `output/` holds `log.csv` (every 100 ticks), `events.json` (collapses, extinctions and fires) and its screenshots. The worlds can also be loaded in the normal Life Engine interface through Load, then World.